

Blanchard and Perotti Model (with Deterministic Trend)

We consider the basic VAR specification as outlined in the paper¹,

$$Y_t = C + \Phi(L)Y_{t-1} + U_t$$

$Y_t \equiv [T_t, G_t, X_t]'$ is a three-dimensional vector in the logarithms of quarterly taxes, spending, and GDP, all in real, per capita terms

$\Phi(L)$ is a four-quarter distributed lag polynomial

$U_t = [t_t, g_t, x_t]'$ - vector of reduced form residuals with nonzero cross correlations

Adding constant and deterministic trend,

$$Y_t = c + D_1t + D_2t^2 + D_3Dummy + S_1q_1 + S_2q_2 + S_3q_3 \\ + A_1Y_{t-1} + A_2Y_{t-2} + A_3Y_{t-3} + A_4Y_{t-4} + U_t$$

$$Y_t = c + D_1t + D_2t^2 + D_3Dummy + S_1q_1 + S_2q_2 + S_3q_3 \\ + A_1Y_{t-1} + A_2Y_{t-2} + A_3Y_{t-3} + A_4Y_{t-4} + A_0^{-1}Be_t$$

where c is a constant and a 3×1 vector; $c = [c_1 \ c_2 \ c_3]'$

- They formalize trends in all three variables as deterministic, and allow for linear and quadratic terms in time (given by t and t^2) in each of the equations of the VAR
- Dummy takes the value of 1 for quarter 1975:2 and for four quarters preceding it
- In order to account for seasonality, q_1 , q_2 and q_3 are included

Identification

$$A_0U_t = Be_t$$

$$\begin{bmatrix} 1 & 0 & -a_1 \\ 0 & 1 & -b_1 \\ -c_1 & -c_2 & 1 \end{bmatrix} \begin{bmatrix} t_t \\ g_t \\ x_t \end{bmatrix} = \begin{bmatrix} 1 & a_2 & 0 \\ b_2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} e_t^t \\ e_t^g \\ e_t^x \end{bmatrix}$$

$$t_t = a_1x_t + a_2e_t^g + e_t^t$$

$$g_t = b_1x_t + b_2e_t^t + e_t^g$$

$$x_t = c_1t_t + c_2g_t + e_t^x$$

- In the literature, this representation of the structural form is often called the AB model

¹Blanchard, O., and Perotti, R. (2002). An empirical characterization of the dynamic effects of changes in government spending and taxes on output. The Quarterly Journal of Economics, 117(4), 1329-1368.

- Without restrictions on the parameters in A_0 and B the structural model is not identified
- This identification scheme relies on a two-step procedure:
 - In the first step, the institutional information is used to estimate cyclically adjusted taxes and government expenditures
 - In the second step, estimates of fiscal policy shocks are obtained.
- $a_1 = 2.08$ - Elasticity of taxes to output; $b_1 = 0$ - Elasticity of spending to output;
- Using $t'_t \equiv t_t - a_1 x_t$ and $g'_t \equiv g_t - b_1 x_t = g_t$ as instruments and getting estimates of c_1 and c_2
- Getting estimate of a_2 by setting $b_2 = 0$
- Thus, the structural spending and tax shocks are identified

$$U_t = A_0^{-1} B e_t$$

Bayesian Estimation

I take a Bayesian perspective to estimate the above reduced-form VAR, treating the model's parameters $\Phi(L)$ and Q as random variables (Bernanke et al., 2005). We further define $vec(\phi)$ as a column vector of the elements of the stacked matrix ϕ of the parameters of the lag operator $\Phi(L)$.

Recalling the VAR specification,

$$Y_t = C + \Phi(L)Y_{t-1} + U_t \quad (1)$$

Further, if we assume p lags and define $Z_t = (1, Y'_{t-1}, \dots, Y'_{t-p})$, then

$$Y_t = Z_t \Phi + U_t \quad (2)$$

Since each equation in the VAR has identical regressors,

$$y = (I_N \otimes Z) \phi + U \quad (3)$$

where $y = vec(Y_t)$, $\phi = vec(\Phi)$ and $U = vec(U_t)$. We now assume that the prior for the VAR coefficients ϕ is normal and given by

$$p(\phi) \sim N(\tilde{\phi}_0, H) \quad (4)$$

where $\tilde{\phi}_0$ is an $[N \times (1 + N \times p)] \times 1$ vector which denotes the prior mean while H is $[N \times (1 + N \times p)] \times [N \times (1 + N \times p)]$ matrix. It can be shown that the posterior distribution of the VAR coefficients conditional on Q is normal (Kadiyala and Karlsson, 1997).

$$h(\phi|Q, y) \sim N(\phi^*, H^*) \quad (5)$$

where

$$\begin{aligned} \phi^* &= H^* \left(H^{-1} \tilde{\phi}_0 + Q^{-1} \otimes Z'_t Z_t \hat{\phi} \right) \\ H^* &= (H^{-1} + Q^{-1} \otimes Z'_t Z_t)^{-1} \end{aligned}$$

$\hat{\phi}$ is an $[N \times (1 + N \times p)] \times 1$ vector of the OLS estimates of the VAR coefficients. The conjugate prior for the VAR covariance matrix is an inverse Wishart distribution with prior scale matrix Q_0 and prior degrees of freedom α_0 .

$$p(Q) \sim iW(Q_0, \alpha_0) \quad (6)$$

Given the prior in equation (8), the posterior for Q conditional on ϕ is also an inverse Wishart distribution.

$$h(Q|\phi, y) \sim iW(Q^*, \alpha^*) \quad (7)$$

where

$$Q^* = Q_0 + (Y_t - Z_t\Phi)'(Y_t - Z_t\Phi) \\ \alpha^* = T + \alpha_0$$

Φ denotes the VAR coefficients reshaped into $(1 + N \times p)$ by N matrix and T is the sample size.

We employ a Minnesota prior for the above VAR as it incorporates the prior belief that the endogenous variables included in the VAR follow a random walk process or an AR(1) process. We set the following prior values:

$$\tilde{\phi}_0 = [0, 0, 0, \dots, 0]'_{N \times (1+N \times p) \times 1}.$$

Prior variance of the distribution of the VAR coefficients, H is given by the following relations for the VAR coefficients ϕ_{bc} :

$$\left(\frac{\lambda_1}{l^{\lambda_3}}\right)^2 \text{ if } b = c \\ \left(\frac{\sigma_b \lambda_1 \lambda_2}{\sigma_c l^{\lambda_3}}\right)^2 \text{ if } b \neq c \\ (\sigma_b \lambda_4)^2 \text{ for the constant and trend components}$$

where b refers to the dependent variable in the b^{th} equation and c to the independent variables in that equation; σ_b^2 is the residual variance of univariate regression of variable b and l is the lag length. λ_1 controls the standard deviation of the prior on own lags. λ_2 controls the standard deviation of the prior on lags of variables other than the dependent variable. λ_3 controls the degree to which coefficients on lags higher than 1 are likely to be zero. $\lambda_4 \rightarrow 0$, the constant terms are shrunk to zero. We set:

$$\lambda_1 = 10; \lambda_2 = 1; \lambda_3 = 0; \lambda_4 = 50$$

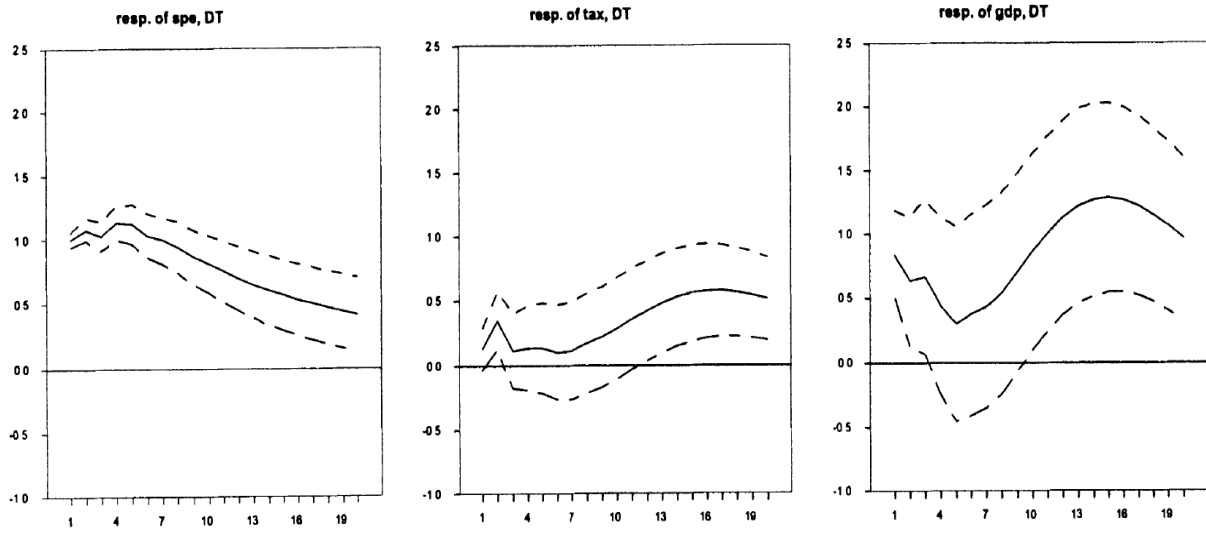
We choose these values to impose a loose Minnesota prior, so that the data play the main role in estimating the VAR. Setting $\lambda_1 = 10$ gives the lag coefficients a large prior variance, making the posterior estimates close to unrestricted OLS. Setting $\lambda_2 = 1$ avoids imposing additional shrinkage on lags of other variables, while $\lambda_3 = 0$ gives the four lags the same degree of prior tightness rather than shrinking higher lags more strongly. Finally, $\lambda_4 = 50$ places a diffuse prior on the constant, seasonal dummies, event dummy, and trend terms. Overall, these choices retain weak Bayesian regularization while keeping the specification close to the frequentist VAR in Blanchard and Perotti (2002).

The initial values of Q and ϕ are taken to be the OLS estimates of the VAR equation. Having set the priors and the initial values, we can now use the Gibbs sampling methodology as follows: Conditional on the value of Q^d , draw ϕ^{d+1} from the conditional posterior distribution of the VAR coefficients. Stationarity is enforced discarding draws of ϕ^{d+1} that contain roots that lie outside the unit circle (Bernanke et al., 2005). Next, we draw Q^{d+1} from its conditional distribution using

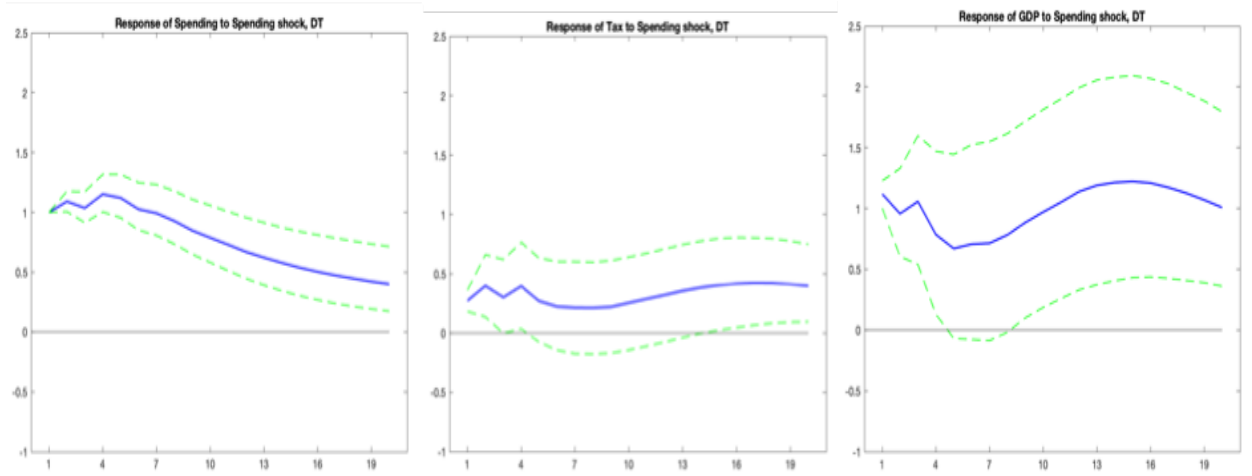
ϕ^{d+1} . These two steps constitute one iteration. For every $(d + 1)^{th}$ iteration, we compute the impulse responses to the spending and tax shocks using the identification technique highlighted above. We use 30,000 Gibbs sampling replications and discard the first 10,000 as burn-in.

Results

Impulse Responses to government spending shock as shown in the paper under the Deterministic model (DT)



Impulse Responses to government spending shock as estimated using the Bayesian approach



To complement the graphical IRFs, the following tables report the responses to spending shock at selected horizons, together with the peak response, to summarize their magnitudes and persistence.

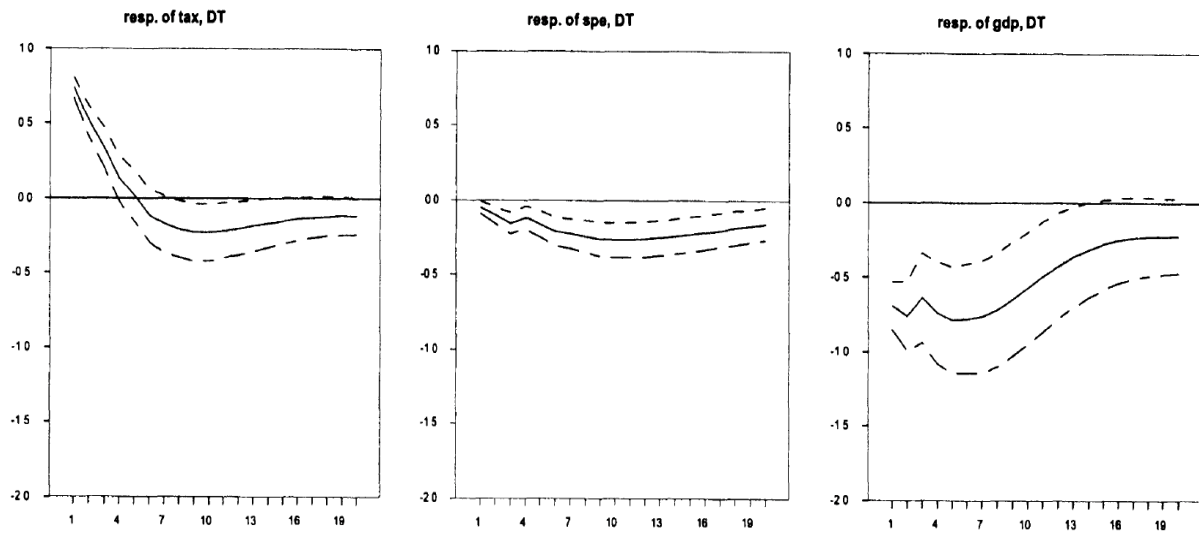
TABLE IV
RESPONSES TO SPENDING SHOCKS

	1 qrt	4 qrts	8 qrts	12 qrts	20 qrts	peak
DT						
GDP	0.84*	0.45	0.54	1.13*	0.97*	1.29* (15)
GCN	1.00*	1.14*	0.95*	0.70*	0.42*	
TAX	0.13	0.14	0.17	0.43*	0.52*	

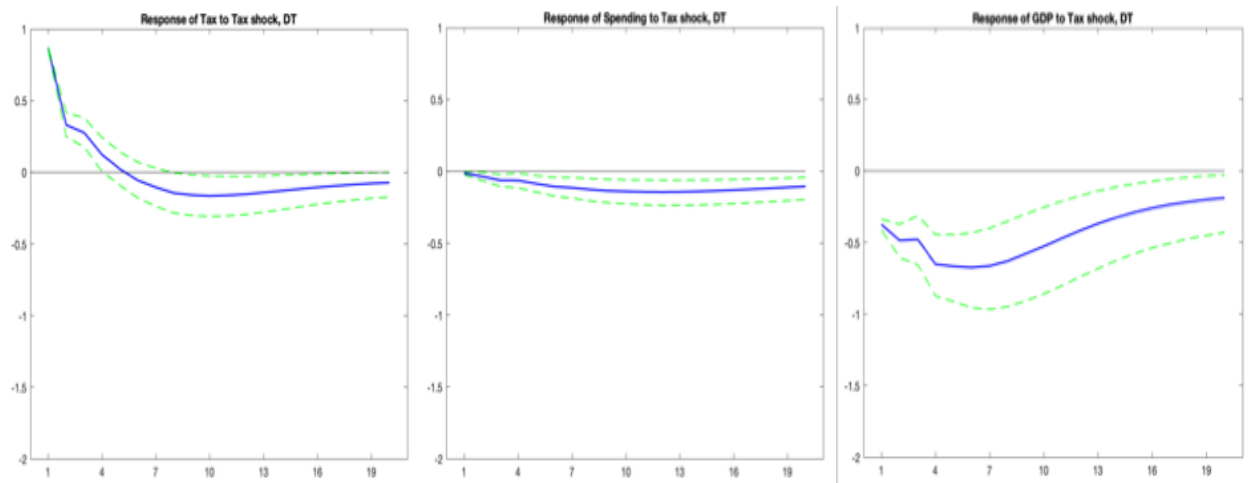
The responses as estimated using the BVAR approach to a spending shock:

	1 qtr	4 qtrs	8qtrs	12 qtrs	20 qtrs	Peak
GDP	1.1*	0.78*	0.78	1.13*	1.0*	1.22* (15)
GCN	1.0*	1.15*	0.93*	0.67*	0.40*	
Tax	0.2*	0.39*	0.21	0.32	0.39*	

Impulse Responses to tax shock as shown in the paper under the Deterministic model (DT)



Impulse Responses to tax shock as estimated using the Bayesian approach



Below are the responses as noted in the paper:

TABLE III
RESPONSES TO TAX SHOCKS

	1 qrt	4 qrts	8 qrts	12 qrts	20 qrts	peak
DT						
GDP	−0.69*	−0.74*	−0.72*	−0.42*	−0.22	−0.78* (5)
TAX	0.74*	0.13	−0.21*	−0.20*	−0.11	
GCN	−0.05*	−0.12*	−0.24*	−0.26*	−0.16*	

The responses as estimated using the BVAR approach to a tax shock:

	1 qtr	4 qtrs	8qtrs	12 qtrs	20 qtrs	Peak
GDP	-0.37*	-0.65*	-0.63*	-0.42*	-0.18*	-0.67* (6)
Tax	0.86*	0.12*	-0.14	-0.15*	-0.07	
GCN	-0.01*	-0.06*	-0.13*	-0.15*	-0.11*	

Note: An asterisk indicates that zero lies outside the 68% posterior credible interval, defined by the 16th and 84th percentiles.

References

- Bernanke, B. S., Boivin, J., & Elias, P. (2005). Measuring the effects of monetary policy: A factor-augmented vector autoregressive (FAVAR) approach. *The Quarterly Journal of Economics*, 120(1), 387–422.
- Blanchard, O., & Perotti, R. (2002). An empirical characterization of the dynamic effects of changes in government spending and taxes on output. *The Quarterly Journal of Economics*, 117(4), 1329–1368.
- Kadiyala, K. R., & Karlsson, S. (1997). Numerical methods for estimation and inference in Bayesian VAR-models. *Journal of Applied Econometrics*, 12(2), 99–132.